

Quick Check

Lesson 1: Anchoring Phenomenon: Echoes at Hell's Gate, Ngong, Kisumu and Nairobi

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Properties of Waves | Lesson 1

Before any teaching, Wanjiru writes down what she thinks an echo is. Which of these is the best everyday description of an echo?

- A** A sound that stays near your mouth and gets louder
- B** A brand new sound that the wall makes by itself
- C** Your own sound coming back to you after it meets a surface
- D** A sound that only the wind can make

The Echo Table shows a clear echo after 0.50 s at the Hell's Gate cliff (85 m away) and a clear echo after 1.00 s at the Ngong Hills valley wall (170 m away). Which statement is supported by these observations?

- A The delay was the same at both places
- B The echo at Ngong Hills came back after a longer delay than at Hell's Gate
- C The echo at Ngong Hills came back after a shorter delay than at Hell's Gate
- D No echo was heard at Ngong Hills

In the Kisumu school hall, the far wall is 12 m away and the delay is 0.07 s. What did the learners record about the sound they heard there?

- A** A faint, clear echo after 0.50 s
- B** No echo at all
- C** A clear echo after 1.00 s
- D** A loud, muddy and drawn-out sound

Learners recorded what they heard at four sites: Hell's Gate cliff, Ngong Hills valley wall, Kisumu school hall and Nairobi school open grounds. Which two sites gave a clear echo?

- A Kisumu school hall and Nairobi school open grounds
- B Ngong Hills valley wall and Kisumu school hall
- C Hell's Gate cliff and Nairobi school open grounds
- D Hell's Gate cliff and Ngong Hills valley wall

In the Echo Table, the Nairobi school open grounds had no hard wall nearer than 300 m and no echo was heard. At this point in the unit, which is the most honest answer to 'Why was no echo heard there?'

- A** Sound disappears when there are no walls close by
- B** The grass swallows all the sound, so nothing can return
- C** The wind always carries the sound away on open ground
- D** We have only recorded what was heard; the reason is still a question we will investigate

The class is building a Driving Question Board about echoes. Which learner question is best for the board because it asks about a cause?

- A** Who wrote the story about the four learners?
- B** Why is there a delay before the echo at Hell's Gate is heard?
- C** In which county is Hell's Gate National Park?
- D** How many learners went on the trip to Hell's Gate?

Amina is drawing her Initial Echo Model of a shout at a cliff. Which set of items should her drawing include?

- A** Only the cliff, because the cliff is the only thing that matters
- B** The person shouting, the cliff, and arrows with labels showing what she thinks travels to the cliff and comes back
- C** Only the person shouting, because the sound stays with the speaker
- D** A final answer with a formula, so that nobody can say the model is wrong

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).